

Internal Audit Report

Claims, Delivery Exceptions and Inventory Adjustment Controls

Transaction-level segregation of duties and fraud risk testing

Entity	Haneul Freight & Logistics (fictional third-party logistics provider, 18 employees)
Audit period	January 2 – March 31, 2026 (Q1 2026)
Report date	April 20, 2026
Overall rating	Needs significant improvement
Prepared by	Minji Kim

This report is part of a personal portfolio project. The company, employees, customers and all records are fictional and were generated as synthetic data. The scenario is inspired by the way roles are commonly combined in small logistics teams; it does not describe any real organization.

1. Executive summary

Haneul Freight & Logistics runs claims, delivery exceptions and inventory adjustments with a small team. Operations Coordinators record shipment status, register and close exceptions, and approve customer claims up to a system-configured limit. In a team this size, combining these duties is a practical necessity. The purpose of this audit was to test whether the controls that are supposed to compensate for that combination actually work.

They do not. Of the ten controls tested, nine were ineffective and one did not exist. The most significant result is a pattern involving one coordinator who, across the full quarter, recorded shipments as lost, closed the related exceptions, and approved the resulting claims without any independent review. The same coordinator raised the claim approval limit fourfold through a system setting, without a change ticket, on a public holiday evening.

These results are **indicators consistent with potential fraud**. They are not a conclusion that fraud occurred. The affected transactions should be referred for further investigation before any conclusion is drawn about intent.

Measure	Result
Records analyzed	2,875 shipments · 11,348 status changes · 194 exceptions · 85 claims · 214 inventory adjustments
Tests performed	11 SQL-based tests (10 control tests, 1 additional analysis)
Controls tested	10 key controls: 9 ineffective, 1 control gap, 0 effective
Findings	3 High · 1 Medium · 1 Low observation
Claims approved without independent review	KRW 11,675,500 (14 claims, one coordinator) – 20.6% of all approved claim value
Claims approved above authorized limit	KRW 8,066,200 (6 claims during an unauthorized configuration change)
Inventory written off without effective approval	KRW 29,531,100 (11 adjustments)

Summary of findings

Ref	Finding	Rating	Amount (KRW)
F1	Single-person control over the loss-claim cycle	High	11,675,500
F2	Unauthorized change to the claim approval limit	High	8,066,200
F3	Inventory write-offs without effective approval	High	29,531,100
F4	Bulk DELIVERED updates without proof of delivery	Medium	–
O1	Low-risk exceptions self-closed by CS staff (observation)	Low	–

F1 and F2 amounts partly overlap; amounts should not be added together.

One control did work: a duplicate claim entered by a customer service associate was rejected by the Operations Manager. The duplicate check operates when the manager approves, and fails when a coordinator approves. That contrast is the clearest illustration of the underlying issue.

2. Background, objectives and scope

Background

When a shipment is lost or damaged, a dispatch coordinator normally records the status change. An exception is registered, the customer submits a claim, and the claim is approved and paid. Claims up to the non-manager limit (KRW 500,000, set in the system parameter CLAIM_APPROVAL_LIMIT_NON_MANAGER) can be approved by an Operations Coordinator; larger claims require the Operations Manager. Separately, warehouse staff post inventory adjustments for damage and count variances, which require approval.

Because the team is small, the Operations Coordinator role can in practice record status, register and close exceptions, and approve claims. Each of those permissions is reasonable on its own. Together, they allow one person to create a loss event and pay for it.

Objectives

- Assess fraud and error risks in the claims, exception and inventory adjustment processes (COSO Principle 8).
- Test whether segregation of duties and approval controls operate in the actual transaction data, not only in the permission design (Principle 10).
- Determine whether IT configuration changes affected business process controls (Principle 11).
- Recommend compensating controls that are realistic for a small team where full segregation is not feasible.

Scope

In scope	Out of scope
Customer claims and approvals · delivery exceptions · shipment status audit trail · proof of delivery records · inventory adjustments · claim-related system configuration history · customer master onboarding records	User access provisioning and general change management (covered in a separate ITGC review) · billing and revenue · carrier invoices · interviews and physical inspection

3. Approach

All tests were performed as SQL queries against the full population of Q1 2026 records, so no sampling was required. Each test is defined by a written rule (R1–R8), a population, and an exception condition, and every query is kept in *queries.sql* so the results can be re-performed.

Two design choices shaped the testing. First, segregation of duties was tested at the transaction level: rather than asking who *could* approve a claim, the tests asked who *actually* recorded, closed and approved each one. Second, the claim approval limit changed during the quarter, so every claim was compared with the limit in effect at its approval time rather than a single fixed value.

Each risk was rated on likelihood and impact (1–5), first as inherent risk, then as current residual risk after considering test results, and finally as target residual risk assuming the recommendations are implemented. The full risk assessment and risk and control matrix are in *Haneul_Risk_Assessment_RCM.xlsx*.

Limitations

Testing was data-only; no interviews, walkthroughs or physical inspection were performed. The after-hours test does not account for public holidays because no holiday calendar table was available. Whether any transaction was fraudulent is outside the scope of this audit and requires investigation.

4. Detailed findings

F1 Single-person control over the loss-claim cycle

Rating: High

Condition

- 14 claims totaling KRW 11,675,500, all paid, were approved by the same Operations Coordinator (E007) who recorded the shipment as LOST. For 11 of them, E007 also submitted the claim. (T1)
- E007 registered and closed 11 LOST exceptions personally, each within 24 hours. (T5)
- 4 of these claims were for shipments that already had a proof of delivery. The status history shows DELIVERED followed days later by a LOST entry from E007. (T4)
- 2 shipments were compensated twice, both times approved by E007 (KRW 945,700). (T3)
- 5 LOST status changes were entered by E007 between 22:00 and 23:00. Every other LOST or DAMAGED entry in the quarter was made during business hours. (T7a)
- 9 of E007's 22 approvals (40.9%) were at 90–100% of the approval limit in effect, compared with 14.3% for the other coordinator. When the limit was raised, E007's claim amounts rose with it. (T2)
- 11 of the 14 claims went to two customers. One of them was onboarded by E007 on December 29, 2025, at 19:47, although customer records are normally created by the CS Lead. (A1)

Criteria

The person who records a loss must not approve the related claim (R1). A shipment may have only one paid claim (R3). LOST claims are invalid where a proof of delivery exists (R4). LOST/DAMAGED exceptions must be closed by someone other than the registrant (R5). COSO 2013 Principles 8 and 10.

Cause

The coordinator role combines recording, exception handling, claim approval and, in practice, customer onboarding. The system does not block the same user from recording a loss and approving the claim, and it does not check for proof of delivery or existing claims before approval. No one reviews claims approved by coordinators.

Effect

One person was able to create a loss event, close the review step, and authorize payment without any independent check. The combination of indicators is consistent with potential fictitious claims. One additional low-value instance (KRW 142,100, a different coordinator) shows the same system gap is available to others.

Recommendation

1. Refer the 14 claims to management for investigation, including confirmation with the two customers and review of the delivered shipments.
2. Configure the system to prevent a user from approving a claim on a shipment where they recorded the LOST or DAMAGED status.
3. Require automatic proof-of-delivery and duplicate-claim checks before any claim can be approved.
4. Introduce a monthly Operations Manager review of all coordinator-approved claims, supported by a report that flags near-limit amounts, after-hours status changes and approver concentration.
5. Restrict customer creation to the CS Lead, with monthly review of new customers.

Owner: Operations Manager

F2 Unauthorized change to the claim approval limit**Rating: High****Condition**

On February 16, 2026 at 22:41, during the Lunar New Year holiday, E007 changed CLAIM_APPROVAL_LIMIT_NON_MANAGER from KRW 500,000 to 2,000,000 without a change ticket or approval (T8a). The setting stayed active for 34 days until the IT Administrator restored it through a ticketed change on March 23. During that window, coordinators approved 6 claims totaling KRW 8,066,200 above the authorized limit: 5 by E007 (KRW 7,386,200) and 1 by E008 (KRW 680,000) (T8b).

Criteria

Every configuration change requires an approved change ticket (R8). Claims above the authorized limit require the Operations Manager (R2). COSO 2013 Principle 11 – general controls over technology.

Cause

An operations role had permission to change system parameters. No ticket was enforced and no alert was raised when an approval parameter changed.

Effect

A single configuration change silently disabled a business process control for more than a month. The E008 approval appears to have been made in good faith under the altered setting, which shows that the impact reached beyond the person who made the change. This is where IT general controls and business process controls meet: a change management failure became an authorization failure.

Recommendation

1. Remove configuration permissions from all operations roles and limit parameter changes to the IT Administrator.
2. Enforce ticket approval for any change to approval limits and thresholds.
3. Send an automatic alert to the Operations Manager and IT when an approval parameter changes.
4. Have the Operations Manager review all six claims approved above the authorized limit.

Owner: IT Administrator

F3 Inventory write-offs without effective approval**Rating: High****Condition**

11 inventory adjustments writing off KRW 29,531,100 of high-value stock were not properly approved (T6): 5 were posted by the Warehouse Lead (E009) with no approval, 3 were approved by E009 himself, and 3 adjustments of KRW 1,000,000 or more, posted by a warehouse associate, were approved by E009 instead of the Operations Manager. Every exception involves E009.

Criteria

Adjustments must be approved by someone other than the adjuster; adjustments of KRW 1,000,000 or more require the Operations Manager (R6). COSO 2013 Principle 10.

Cause

The system allows adjustments to post without approval, accepts self-approval, and does not route high-value adjustments to the Operations Manager.

Effect

Write-offs are a common way to conceal inventory loss or theft. The recorded inventory may also be misstated. The concentration on one person and on high-value items raises the priority of this finding.

Recommendation

1. Physically verify the stock position for the 11 affected items.
2. Require approval before an adjustment posts, block self-approval, and route adjustments above the threshold to the Operations Manager automatically.
3. Assign periodic cycle counts of high-value items to someone outside the warehouse team.

Owner: Operations Manager

F4 Bulk DELIVERED updates without proof of delivery

Rating: Medium

Condition

On March 31, 2026, the last day of the quarter, the Operations Coordinator E008 set 28 shipments to DELIVERED within eight minutes (18:52 to 18:59). None has a proof of delivery record. Every other DELIVERED status in the quarter is supported by a POD (T7b).

Criteria

A DELIVERED status must be supported by a proof of delivery record (R7). COSO 2013 Principle 13 – quality information.

Cause

The system allows DELIVERED status without a POD. The timing suggests pressure to meet a quarter-end delivery metric.

Effect

No money was paid out, and E008 does not appear in the claim pattern in F1. The risk is to the reliability of operational reporting: on-time delivery rates may be overstated, and customers may be told goods were delivered when they were not. Not every exception is fraud, and this one should be reported as a reporting-integrity issue.

Recommendation

1. Confirm the actual delivery status of the 28 shipments.
2. Require a POD record before a shipment can move to DELIVERED.
3. Include a DELIVERED-without-POD report in the quarter-end KPI sign-off.

Owner: Dispatch / Operations Manager

O1 Low-risk exceptions self-closed by customer service staff (observation)

Rating: Low

Condition

Six DELAY and ADDRESS_CORRECTION exceptions were registered and closed by the same customer service associate within a few hours (T5).

Criteria

R5 requires independent closure for LOST and DAMAGED exceptions only. The policy is silent on other types.

Cause

No written policy defines which exception types can be self-closed.

Effect

These exceptions do not lead to payouts, so the risk is low. Requiring independent review for them would add work without reducing meaningful risk.

Recommendation

1. Document that DELAY and ADDRESS_CORRECTION exceptions may be self-closed, and keep independent closure mandatory for LOST and DAMAGED.

Owner: CS Lead

5. Compensating controls for a small team

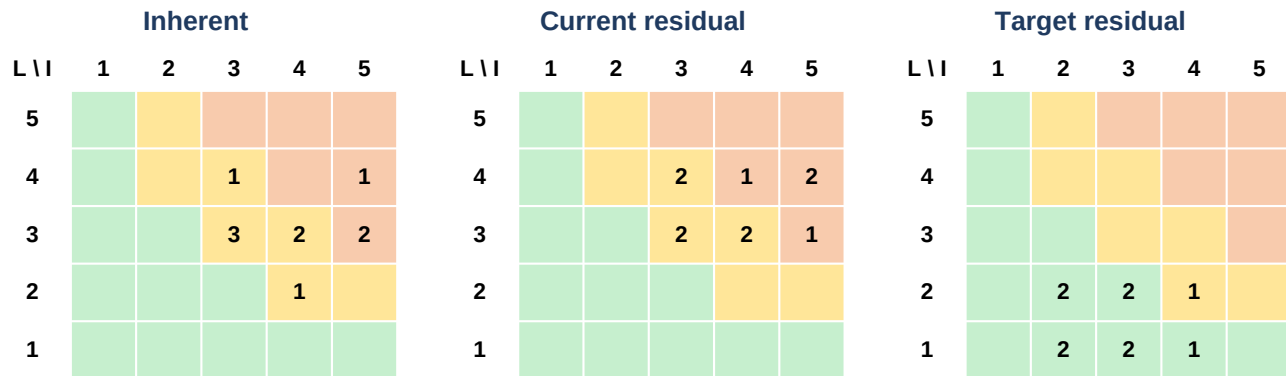
Hiring more staff to fully separate these duties is not realistic for an 18-person operation. The recommendations above are designed so that one person can still perform several roles, as long as the system blocks the most dangerous combinations and someone independent looks at the results regularly.

Duties that stay combined	Compensating control	Type	Frequency	Effort
Record LOST/DAMAGED + approve claim	System block on same-user approval for the same shipment	Preventive, automated	Every transaction	Low (configuration)
Approve claims up to the limit	Manager review of a claims report flagging near-limit amounts, approver concentration and repeat customers	Detective, manual	Monthly	Medium
Change status at any time	Exception report of LOST/DAMAGED changes outside business hours or by non-dispatch staff	Detective, manual	Weekly	Low
Handle exceptions end to end	Independent closure required for LOST/DAMAGED only; low-risk types may be self-closed	Preventive, manual	Every transaction	Low
Post and approve inventory adjustments	Approval-before-posting workflow, self-approval block, automatic routing above threshold	Preventive, automated	Every transaction	Low (configuration)
Operations staff with system access	Configuration rights limited to IT; alert on any change to approval parameters	Preventive + detective	Every change	Low
Quarter-end delivery reporting	DELIVERED-without-POD report included in KPI sign-off	Detective, manual	Quarterly	Low

Most of these controls are system settings or short reports that can be generated from the same queries used in this audit. The ongoing effort is roughly a few hours per month for the Operations Manager.

6. Risk assessment summary

Heat maps show the number of risks in each cell (L = likelihood, I = impact). Risks move up and right from inherent to current residual because the key controls failed, and fall to the lower left once the recommendations are in place.



Risk	Description	Inherent	Current residual	Target residual
R-01	Fictitious claim recorded and approved by one person	High (20)	High (20)	Low (6)
R-02	Approval limit circumvented by structuring amounts	Medium (12)	High (16)	Low (6)
R-03	Duplicate claim paid	Medium (9)	Medium (9)	Low (2)
R-04	LOST claim paid on a delivered shipment	Medium (12)	Medium (12)	Low (3)
R-05	Exception closed without independent review	Medium (12)	Medium (12)	Low (4)
R-06	Write-offs conceal inventory loss or theft	High (15)	High (20)	Medium (8)
R-07	Status changed outside hours and oversight	Medium (9)	Medium (9)	Low (4)
R-08	Delivery performance overstated	Medium (9)	Medium (12)	Low (2)
R-09	Configuration change disables a business control	High (15)	High (15)	Low (4)
R-10	Customer created by staff who process its claims	Medium (8)	Medium (12)	Low (3)

Score = likelihood × impact (1–5 each). High ≥ 15, Medium ≥ 8. Current residual risk is at or above inherent risk for every item because the primary controls failed testing. Target residual assumes the recommendations are implemented.

Appendix A. Test results

Test	Rule	Description	Population	Exceptions	Rate
T1	R1	Loss recorder approved the claim	85 claims	15	17.6%
T2	R2	Approval at 90–100% of limit in effect	36 approvals	11	30.6%
T3	R3	Second paid claim on the same shipment	82 shipments	2	2.4%
T4	R4	LOST claim with proof of delivery	43 LOST claims	4	9.3%
T5	R5	LOST/DAMAGED exception self-closed within 24h	192 exceptions	11	5.7%
T6	R6	Adjustment unapproved, self-approved or under-authorized	214 adjustments	11	5.1%
T7a	R7	LOST/DAMAGED status changed outside business hours	91 status records	5	5.5%
T7b	R7	DELIVERED status without proof of delivery	2,679 status records	28	1.0%
T8a	R8	Configuration change without approved ticket	5 changes	1	20.0%
T8b	R2, R8	Approval above authorized limit during unauthorized window	16 approvals	6	37.5%
A1	–	Flagged customer onboarded by an involved employee	5 customers	1	20.0%

T3 found three shipments with duplicate claims; one duplicate was rejected by the Operations Manager and is excluded from exceptions.
T5 returned 17 records; six low-risk self-closures are reported as observation O1.

Appendix B. Business rules used as audit criteria

Rule	Criterion
R1	The person who records a shipment as LOST or DAMAGED must not approve the related claim.
R2	Non-manager claim approvals are limited to the authorized value of CLAIM_APPROVAL_LIMIT_NON_MANAGER (KRW 500,000).
R3	A shipment may have only one paid claim.
R4	A LOST claim is invalid if a proof of delivery exists.
R5	LOST and DAMAGED exceptions must be closed by someone other than the registrant.
R6	Inventory adjustments require an independent approver; KRW 1,000,000 or more requires the Operations Manager.
R7	Business hours are Mon–Fri 08:00–19:00. DELIVERED status must be supported by a proof of delivery.
R8	Every system configuration change must have an approved change ticket.

Appendix C. Deliverables

haneul_logistics_audit.db (synthetic SQLite database) · queries.sql (test queries T1–T10) ·

Haneul_Risk_Assessment_RCM.xlsx (risk assessment, heat maps, risk and control matrix, test results, findings log) · this report. All data is synthetic; no real company data was used.